

Peer to Peer Technologies

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1 PEER TO PEER (P2P)

1.1 Zoom

Peer to peer applications nowadays are abundantly excessive in society's usual work and life. One of the most popular peer to peer applications that almost everyone has used is Zoom. As you may know, Zoom is a video and telephone calling software that was widely used a few years ago when the pandemic was causing chaos. The introduction of this application was the beginning of a massive transition towards predominately p2p applications around the world. Before the necessity of Zoom for work, p2p applications were used quite often. However, Zoom was the application that introduced and solidified the change in how parts of the society's work life has become hybrid or work from home. Zoom's main functionality is the use of video and phone calls online with other people. However, it also offers a wide variety of other functionalities such as screen share and file share. Similarly to Zoom, there are many other p2p applications that offer very similar functionalities such as Skype and Google Hangout. Ultimately though, Zoom was the technology that caused the greatest technological impact around the world. Because of this, the use of p2p applications for communication and work has become a complete norm even though many people are able to work in offices once again. Zoom has completely redefined the way workers are able to work at home and in offices.



Figure 1—Zoom logo. [Source](#)

1.2 Insights on Zoom

Like many other people working online throughout the pandemic, I became fairly familiar with Zoom since I used the application regularly. From my understanding of the application, there are many reasons why Zoom was a good and bad application. The main reason Zoom can be considered a good application and was clearly able to surpass similar applications during the

time was its clear adaptability towards certain situations and its scalability. At the time, users needed an application that offered the most reliable video calling service that can handle the most amount of users reliably. In this case, Zoom was quickly able to identify this and match its service to what consumers needed. In addition, it was able to foresee that user's would need the ability for screen sharing and screen controlling during these video calls and was able to implement these functionalities well. These aspects allowed Zoom to climb the ranks as one of the best applications on the market. On the other hand, although Zoom was able to climb quickly, it has since then fallen off the market as other applications such as Microsoft Teams and Skype take over once again. This can be seen through Zoom's stagnant development. Although Zoom arguably has the best technology in video calling, other applications offer a wider variety of technologies in addition to video calling functionality, such as IMS, calendars, group chats, etc. Without the continued addition of other functionalities within the application, Zoom simply will not be able to hold its own with other applications that offer more.

1.3 Centralized vs Decentralized Coordination

	Centralized Coordination	Decentralized Coordination
Advantages	• Single point of control • Central data storage • Controls all communication from one point	• Multiple control points so there are more than one point with authority • Data distribution across multiple points • Redundancy against lost data • Split authority so no point has complete control
Disadvantages	• Single point of failure	• Not as scalable as centralized

2 P2P OVERLAY NETWORKS

2.1 Advantages/Disadvantages

Structured peer to peer systems and unstructured peer to peer systems both have their pros and cons. First and foremost the search protocol of the structured system and the unstructured system are completely different due to the topological structure of the graph. With a formatted structure, the search protocol for this type of peer to peer system would be deterministic, which means that

the results of the search are reproducible and predictable. The search protocol will likely give the same answer every time. However, the unstructured peer to peer system will be opportunistic, which means that the results are dependent on the information encountered during the search as well. Performance wise, unstructured p2p would be better since opportunistic search is geared towards performance. Correctness is more towards deterministic structure which is structured p2p. Data placement of unstructured and structured systems also differ. Unstructured systems have data shared in a decentralized fashion with local placement while structured systems have a global placement of data that is then distributed. With two different types of data placement, performance can differ depending on what you are looking for. If you want a more efficient and simplistic system then a structured system is better. However, if you are looking for a more resilient and secure system then an unstructured system is overall better.

2.2 Youtube

I believe that Youtube is a platform that could ultimately benefit from adding in peer to peer technology in the future. Youtube currently does not have any peer to peer technology within its use. However, Youtube has slowly begun to move into the field of streaming with many major streamers using Youtube as their main streaming platform. Other streaming sites have already been using peer to peer technology on their streaming sites for a few years now and it has been determined the peer to peer live streaming were able to offer high definition content reliably and consistently. Since Youtube is a strong contender in the field of live streaming, I believe that the addition of peer to peer technology can improve the content quality that streamers on Youtube are currently offering.

2.3 Design/Implement

Peer to peer streaming is still a fairly new concept that has only been implemented in only a few streaming sites. My thought process of adding this implementation to Youtube is very similar to what other sites have today. Although Youtube's video streaming site utilizes a centralized point for all its video accessing and playing, the live streaming section of Youtube will utilize an unstructured peer to peer system. Since the live stream users itself will act as the distribution nodes, there will not be any need for additional hardware (P2P Live Streaming: Transform Your Viewing Experience Today!, 2024). Since the quality of the streams are dependent on the distribution nodes active, the quality of the videos will be linked towards the number of users. This can guarantee that high traffic streamers will have good video quality. In addition, there can be a system in play to allow peer to peer systems not be in use unless a stream has a certain number of viewers to ensure a certain level of quality of the site itself.

3 LOCATION BASED SERVICES

3.1 Yelp/Waze

Waze is a mobile application that has location acquisition services that I tend to use on a regular basis. Waze utilizes location based navigation and location based traffic monitoring to advise the user of navigation and traffic conditions on the road. Waze is an excellent technology since it utilizes live updates to the road which can give the user's navigation real time update on the fastest ways to get to his or her destination. In addition, this real time update can be helpful with traffic monitoring as well. On the other hand, traffic updating on the navigation location service can be quite inaccurate. The application utilizes user reviews and input based on his or her location for updates to traffic. This causes traffic change on the navigation to have higher latency than opposing location acquisition services.

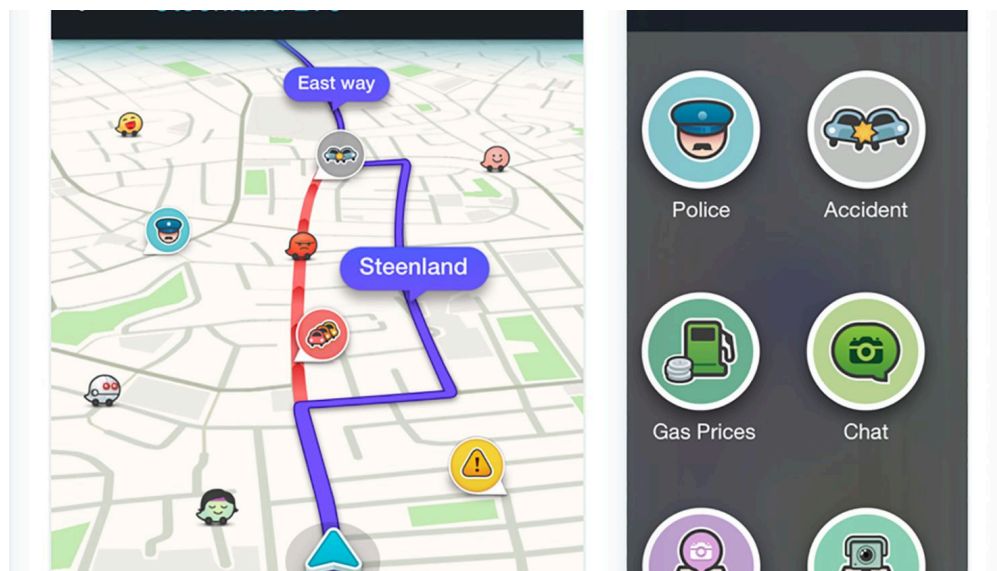


Figure 2—Waze GPS Map and user traffic input. [Source](#)

3.2 Redesign

If I were to redesign this specific location based mobile service, I would focus heavily on improving the traffic monitoring aspect of the location based service. The app currently relies too heavily on user input to determine police, traffic, and accidents. The service needs to be able to extrapolate from the user's location and the user's inputs to determine what situation traffic is in. In addition, there needs to be a higher screening of the accuracy in what users are experiencing on the road. A possible way of assisting with both of these could be the inclusion of video data from front and back dashcams. If user's are willing to accept the usage of this footage to assist with traffic then user error in inputs can be removed all together and the footage can ultimately be used in conjunction with user location to determine traffic conditions on the road.

3.3 GPS Positioning

	Pros	Cons
GPS Positioning	• Extreme accuracy of target when available • Most effective in areas of less structures	• Situational accuracy where signal can be blocked in doors • Slower than cellular and wifi
Cellular network positioning	• Most effective in dense populated area of cell phone towers	• Depends on cell phone towers in the area • Require access to a database of ID locations
Wifi Positioning	• Utilize wireless access points for strength • Accuracy up to a few meters	• Reliant on WiFi access points • Require access to a database of ID locations

4 FUTURE INTERNET SYSTEMS AND SERVICES

4.1 Traveling on the Road

As a fellow driver who is often on the road as well, I believe that out of all the different types of location based services, mapping, navigation, and location are the most important. Traveling on the road can be very dangerous, especially when you are traveling to unknown places. Location based services slowly become less important the more familiar you are traveling to a certain place, so in this hypothetical situation I am assuming that you are traveling to a far and unusual place. In this scenario, location easily becomes the most important aspect of location services needed. Without being able to know your own general location, no other location services will be of help at all. The next location service that would be most important is mapping. As long as you have your own general location and a map of the surrounding area your chances of survival increases significantly. As long as you are able to somewhat decipher the map and determine a logical direction to go in, you are almost guaranteed survival. Finally, the last location based services that I have chosen to be the most helpful would be navigation. With your own general location, a mapping of your surroundings, and navigation towards your intended location, it is highly unlikely that anything alarming can happen on the road. These three location based services together make up the crutch of any GPS application that can assist a person from driving from point A to point B.

4.2 Experiences

Mobile location based services are a highly important aspect in my daily life that I usually take for granted. These services are almost always an assistance whenever I leave the house. Driving the car, making a phone call, checking destination locations, and even sitting in class, all have some aspect that is improved through mobile location based services. All the applications and functionalities that I have in my phone such as GPS look up, GPS navigation, wifi, and cellular networks all assist in my everyday life to help improve on certain aspects. However, although mobile location based services are very helpful overall, there are still some aspects that can be further improved and can be a hindrance in my everyday life. For example, data privacy is something that I personally take very seriously, especially since I know how easy it is to steal someone else's data for malicious use. That is why I always try to ensure that my personal data is protected. However, with how abundant data is constantly being collected by mobile location based services, it is difficult to fully determine whether or not what data is being taken and for what purpose. I believe a possible way to improve this is higher transparency on what data is being collected. I know that currently you can check on what data is being recorded on apps if you go through settings on a phone to find it. However, these convoluted steps cause many people to get lazy and ignore this. I believe that if the services are set to remind the user before using or collecting personal data then the user must be aware of what data is being collected, and if the user is given the option of never see the prompt again then this can clear up a lot of privacy data issues where data is being taken without the user's knowledge.

4.3 Scalable Location-Based Advertisement

When trying to develop a scalable advertisement system based on location, it is very important to be able to be conscious of how the location will change the advertisement. This opens up many different avenues of how each location will determine the advertisement. I believe that the actual advertisement split based on location should be the main aspect to determine before anything else since it will determine the actual size of the database used. Personally I believe that since the U.S. is already split into specific counties already, we can utilize this to our advantage and determine a set number of advertisements based on the user's location in what county to give predetermined advertisements. The advertisements themselves will be split based on partnered companies and time slots bought. This will cut out the most difficult part of the algorithm and leave the location services to determine the county for the advertisements to be displayed.

5 REFERENCES

1. *P2P Live Streaming: Transform Your Viewing Experience Today!* (2024, May 9). StreamShark. <https://streamshark.io/enterprise/peer-to-peer-live-streaming/>